

KeralamMatch

End-to-End System Flowchart & Architecture Documentation

Platform Specification & Technology Roadmap — Version 1.0

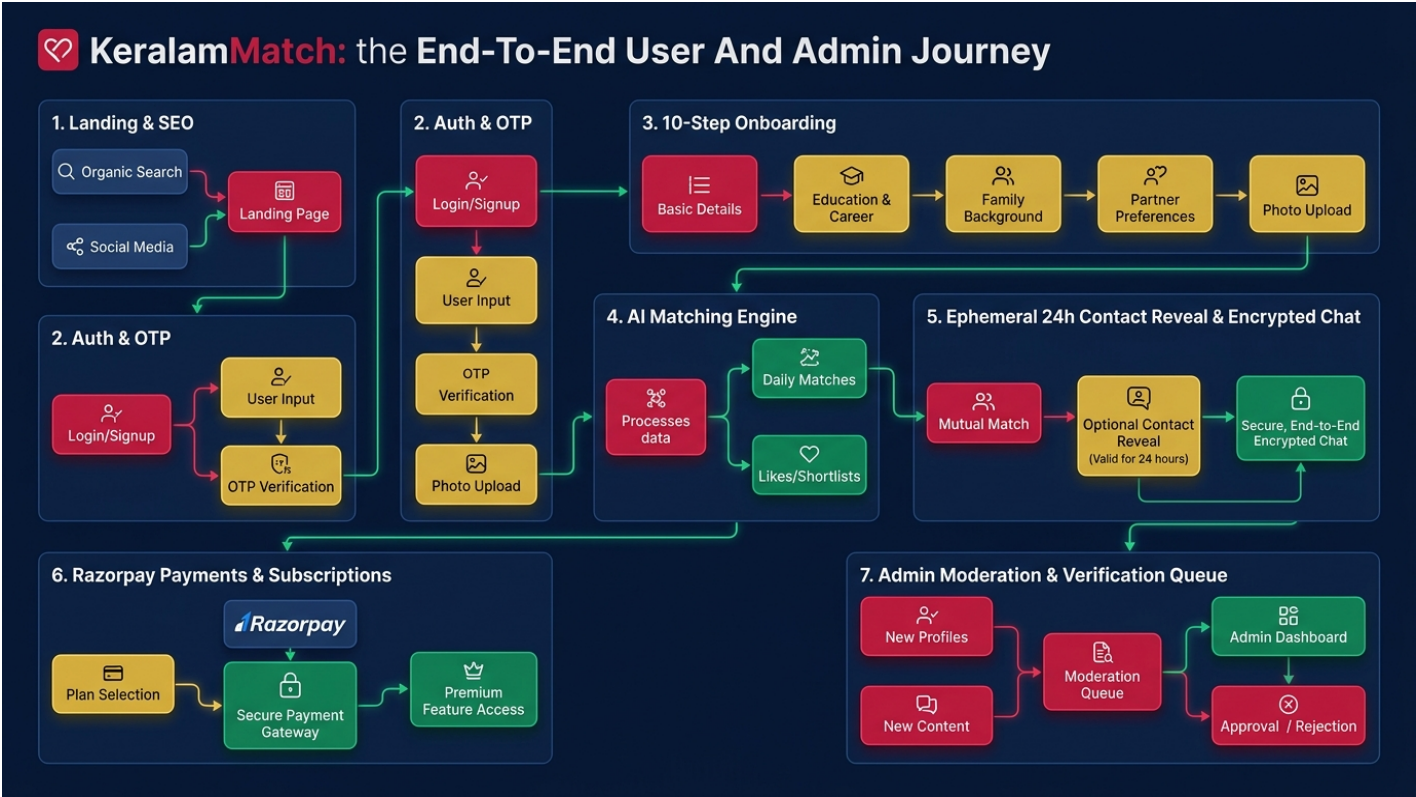


Figure 1: KeralamMatch End-to-End User and Admin Journey Flowchart Diagram

1. Executive Summary & Vision

KeralamMatch is Kerala's premier privacy-first, verified matrimonial platform designed for the global Malayali community. Unlike traditional matrimonial portals that publicly expose candidate phone numbers, photos, and personal details to search engines or anonymous web crawlers, KeralamMatch enforces consent-based 24-hour ephemeral contact reveals, AES-256-GCM database field encryption, multi-layer identity verification, and AI-assisted attribute compatibility matching.

2. Platforms & Technology Stack (Used Now)

Domain / Layer	Platform / Tech Used	Responsibility & Function
Frontend Framework	Next.js 16 (App Router) + React 19	Server/client rendering, SSG for 53 routes, Turbopack engine
Language	TypeScript (Strict Mode)	End-to-end type safety, schemas, and repository interfaces
Design System	Tailwind CSS v4 + Lucide Icons	Custom design tokens (#0A1F44, #C81D45, #D4AF37, #FCFBF7)
Database Layer	Neon PostgreSQL + Prisma ORM v6	Serverless PostgreSQL with SSL encryption, model relations
Data Security	AES-256-GCM Field Encryption	Encrypts phone numbers & emails at rest before DB insertion

Authentication	Firestore Auth (SMS OTP + Google)	Mobile OTP verification & encrypted session cookie (km_session)
Media Cloud Storage	Cloudinary CDN	AVIF/WebP image optimization, watermarking, liveness check
Payment Gateway	Razorpay API + Webhooks	UPI, Cards, Netbanking checkout for membership plans & credits
Email Infrastructure	Resend API	Transactional HTML emails for OTP, requests, and receipts
Security & Caching	Upstash Redis + Security CSP	Rate-limiting Auth/Chat APIs, HSTS, and CSP header injection
Analytics	Google Analytics 4 + Clarity	Traffic metrics, event tracking, and user session recording

3. Future Platforms & Integration Roadmap (Going to be Used)

- **Native Mobile Apps (React Native / Expo)**: Native apps for Apple App Store & Google Play Store featuring FaceID/Biometrics login and camera verification.
- **DigiLocker Govt Verification (SurePass / Cashfree API)**: Automated 5-second instant Aadhaar, Passport, and PAN verification to issue green verified badges.
- **WhatsApp Business API (Twilio / Meta WhatsApp API)**: Deliver instant WhatsApp alerts for incoming contact requests, chat messages, and 24h reveal expiry warnings.
- **AI Vector Matcher Engine (OpenAI Embeddings + Pinecone)**: Hyper-personalized semantic candidate matching based on vector embeddings and Porutham (horoscope matching).
- **In-App Virtual Video Dates (Agora.io / Daily.co WebRTC SDK)**: Encrypted 1-on-1 virtual video dating directly inside the platform without disclosing personal phone numbers.
- **Push Notification Service (Firebase Cloud Messaging (FCM))**: Real-time web push and mobile push notifications for instant candidate activity alerts.

4. End-to-End User & Admin Flow Stages

- **Stage 1: Entry & SEO**: Visitors arrive via main Landing Page (/), District SEO (/find/brides-in-district), Community pages (/find/nair-matrimony), or Blog (/blog). Masked profiles rendered for guests.
- **Stage 2: Auth & 10-Step Onboarding**: User logs in via Mobile OTP (/auth), receives km_session cookie, and completes 10 onboarding steps (/join): Basic Details, Physical Attributes, Religion/Caste, Education, Location, Family, Lifestyle, Partner Preferences, Cloudinary Photos, and Liveness Check.
- **Stage 3: Discovery & AI Matching**: Member views Dashboard (/dashboard) and searches candidate profiles (/find) using quick filters or AI prompts. System calculates Compatibility Score (%) across 12 key attributes.
- **Stage 4: Ephemeral 24-Hour Contact Reveal**: Candidate A requests contact (/requests). Candidate B receives notification and approves. System unlocks decrypted phone & email for exactly 24 hours (expiresAt = now + 24h). Credentials auto-lock after 24h.
- **Stage 5: Secure Messaging & Monetization**: Matched candidates chat in real-time (/chat). Users upgrade membership plans (Silver/Gold/Platinum) or buy 99 reveal credits via Razorpay modal (/pricing).
- **Stage 6: Governance & Admin Portal**: Admins authenticate (/admin/login) and manage Dashboard stats (/admin), User Management (/admin/users), Verification Queue approvals (/admin/verification), Report Moderation (/admin/reports), Payments (/admin/payments), Blog CMS (/admin/blog), FAQ (/admin/faq), Audit Logs (/admin/audit), and System Settings (/admin/settings).